(i)

$$\because n > 1,\ k \geq 1$$

$$\therefore (k+1)n^{k+1} > kn^k$$

$$\frac{1}{(k+1)n^{k+1}} < \frac{1}{kn^k}$$

When $x = \dfrac{1}{n}$,

$$\ln\left(1+\frac{1}{n}\right) = \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \frac{1}{4n^4} + \cdots$$

$$\begin{aligned}\ln\left(1+\frac{1}{n}\right) &= \frac{1}{n} - \left(\frac{1}{2n^2} - \frac{1}{3n^3}\right) - \left(\frac{1}{4n^4} - \frac{1}{5n^5}\right) - \cdots \\ &< \frac{1}{n}\end{aligned}$$

$$\implies \ln\left(1+\frac{1}{n}\right)^n < 1$$

$$\left(1+\frac{1}{n}\right)^n < e$$

(ii) Let $u = \dfrac{1}{2y},\ y > \dfrac{1}{2} \implies 0 < u < 1$

$$\begin{aligned}\ln\left(\frac{2y+1}{2y-1}\right) &= \ln\left(\frac{1+u}{1-u}\right) \\ &= \ln(1+u) - \ln(1-u) \\ &= \left(u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots\right) - \left(-u - \frac{u^2}{2} - \frac{u^3}{3} - \cdots\right) \\ &= 2\left(u + \frac{u^3}{3} + \frac{u^5}{5} + \cdots\right) > \frac{1}{y}\end{aligned}$$

When $y = n + \dfrac{1}{2}$

$$\frac{2y+1}{2y-1} = \frac{2n+2}{2n} = 1 + \frac{1}{n}$$

$$\ln\left(1+\frac{1}{n}\right) > \frac{1}{n+\frac{1}{2}}$$

$$\implies e < \left(1+\frac{1}{n}\right)^{n+\frac{1}{2}}$$

(iii)

$$e < \left(1+\frac{1}{n}\right)^{n+\frac{1}{2}} = \left(1+\frac{1}{n}\right)^{n}\sqrt{1+\frac{1}{n}}$$

$$\frac{e}{\sqrt{1+\frac{1}{n}}} < \left(1+\frac{1}{n}\right)^{n} < e \qquad \text{Parts (i)(ii)}$$

$$\lim_{n\to\infty}\frac{e}{\sqrt{1+\frac{1}{n}}} = e < \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n} < e$$

$$\therefore \left(1+\frac{1}{n}\right)^{n} \to e$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: July 2, 2026